CNG SmartPlan

Data Package Guide and One Implementation Plan

PS000002 - Station-Level Demand Forecasting for CNG Sales to Optimize Network Expansion

9 CSV files	2 Python scripts	1 ZIP data package
Real public records are partial	3 datasets are synthetic	Coordinates still unverified

The key truth about this package

This is a data-preparation package for a prototype, not a complete operational CNG dataset. It combines partially verified public station-directory records, a small set of unverified coordinate leads, historical/public infrastructure context, documentation, and three Python-simulated time series. No actual Gujarat Gas station-level sales or supply records were obtained.

AI-GENERATED DATA? None of the factual station records or infrastructure indicators were generated by an AI model. The numeric sales, supply, and hourly-demand records are synthetic values produced by a fixed-seed Python simulation. They are not observations and not AI/ML forecasts.

Prepared from the current CNG_SmartPlan_Data package and its validation report. Counts below are from the files checked on 29 October 2026.

1. Data classifications and one clear plan

Use these labels consistently in your website, demo, report, and presentation.

Label	Files / examples	Meaning
REAL PUBLIC - PARTIALLY VERIFIED	cng_stations.csv; optional/cng_infrastructure.csv	Names/addresses were extracted from operator directories; infrastructure indicators were taken from PPAC. These are not a complete station census, and not all operating facts or coordinates are independently verified.
SECONDARY COORDINATE LEADS	station_coordinate_candidates.csv	GPS points came from CNGbunk secondary pages. All are explicitly unapproved for production map pins and require independent checks.
SYNTHETIC / PYTHON-SIMULATED	synthetic_daily_sales.csv; synthetic_supply.csv; synthetic_hourly_demand.csv	Generated from assumed profiles, weekday/season rules, random variation, and a fixed random seed. These files are for workflow demonstrations only.
METADATA / DOCUMENTATION	data_sources.csv; data_dictionary.csv; data_gaps.csv; README.md; validation_report.md	Describe source provenance, column meanings, gaps, usage, and checks. They are not measurements of CNG operations.

One implementation plan

Stage	Action
1. Verify the station master	Resolve station identity, CNG availability, operator/brand discrepancies, and city-boundary inclusion. Obtain authoritative coordinates; do not use candidate points as map pins yet.
2. Prototype the forecast pipeline	Use synthetic_daily_sales.csv to test import, chronological train/test split, baseline comparison, and 7-day forecast output. Label all results as synthetic-demo.
3. Test supply-alert logic	Join synthetic forecast output to planned supply by station and date. Calculate max(0, forecast demand - planned supply); label it a potential planning shortfall.
4. Enable map and impact simulation	Only after coordinates and boundary are verified, calculate distances. Make demand capture, diversion, and accessibility assumptions adjustable and visible.
5. Validate with real operations data	Request authorized station-level sales and supply history from Gujarat Gas/other operators. Evaluate on a chronological holdout before making real-world accuracy claims.

Do not delay frontend data-ingestion or model-pipeline experiments if you need a demonstration. However, the complete map-based planning feature is blocked by missing verified coordinates and incomplete coverage.

2. Station and coordinate CSVs

cng_stations.csv

Classification: REAL PUBLIC - PARTIALLY VERIFIED

Purpose: Master table of station names and address-level information from the two operator directories below. 12 rows x 22 columns; 11 locality-screened rows are eligible for synthetic demonstrations, while 1 is marked uncertain/excluded.

Source websites: Gujarat Energy Limited (formerly Gujarat Gas Limited) CNG Stations directory - https://www.gujarat-energy.com/cng/cng-stations/ (two Chandkheda rows); Hindustan Petroleum Corporation Limited (HPCL) CNG outlet directory - https://www.hindustanpetroleum.com/pages/cng (ten Ahmedabad-address entries).

Columns: station_id, station_name, address, locality, city, district, state, pincode, latitude, longitude, operator, brand, cng_status, station_source_url, coordinates_source_url, verification_method, coordinates_verified, verification_status, in_study_area, source_access_date, data_type, verification_notes

Use: Stable station IDs join the station-level datasets. Source URLs and verification notes preserve provenance.

Important limitation: Latitude and longitude are blank for all rows. The list is a sourced subset, not a complete cross-operator census. Directory listing does not prove a station was physically inspected or is currently operating.

station_coordinate_candidates.csv

Classification: SECONDARY RESEARCH LEADS - NOT VERIFIED

Purpose: 8 rows x 12 columns containing possible coordinates associated with a subset of station IDs. These points are kept separate from the main station master.

Source website: CNGbunk.com secondary station detail pages - https://cngbunk.com/. Individual source URLs are recorded row by row in the CSV and in data_sources.csv.

Columns: station_id, station_name, candidate_latitude, candidate_longitude, candidate_coordinate_source_url, candidate_coordinate_source_name, source_claim, independent_check_status, cng_status_conflict, approved_for_map_pin, data_origin, notes

Use: A review queue for matching a candidate coordinate to an authoritative operator/map record and a validated municipal boundary.

Important limitation: Every record has approved_for_map_pin=no. Some pages raise uncertainty about whether a CNG dispenser is present. Two candidate pages label Adani CNG while their primary directory entries came from Gujarat Energy; resolve that conflict before using them.

Station readiness decision

Do not use coordinate candidates to calculate nearest-station distance, mark stations as verified on a public map, or rank new locations until each coordinate and station identity has an independent supporting source. The candidate coordinates are not silently copied into cng_stations.csv.

3. The three synthetic time-series CSVs

These files are generated by generate_synthetic_data.py. Their data_type column is synthetic. The fixed seed is 20241029, allowing the same inputs and rules to reproduce the same values.

synthetic_daily_sales.csv

Classification: SYNTHETIC - PYTHON SIMULATION

Purpose: 8,041 rows x 12 columns. One simulated daily quantity per eligible station and date from 1 January 2024 through 31 December 2025. 11 station IDs x 731 days = 8,041 rows.

Columns: station_id, date, synthetic_sales_kg, day_of_week, is_weekend, month, year, is_public_holiday, event_name, simulation_profile, data_type, generation_method

Generation logic: assumed station profile base levels, weekday multipliers, a mild annual seasonal cycle, a small drift, and controlled random variation. Holiday/event fields remain unassigned unless a source-backed holiday file is added.

Why synthetic? Real station-day sales data was not available from public sources; these values let the team exercise data ingestion and forecasting workflows.

Important limitation: The values are not actual sales at the named stations, not an AI-generated truth, and not an AI/ML forecast.

synthetic_supply.csv

Classification: SYNTHETIC - PYTHON SIMULATION

Purpose: 8,041 rows x 5 columns. One assumed planned available supply quantity for every station/date key in the daily dataset.

Columns: station_id, date, planned_available_supply_kg, data_type, generation_method

Generation logic: random supply factors create some simulated adequate days and some days where assumed supply is below simulated sales.

Why synthetic? Actual station inventory, deliveries, and supply plans were not obtained; assumed values support a demo of alert logic.

Important limitation: Do not describe this as actual stock, delivery, dispensing capacity, or a Gujarat Gas supply plan.

synthetic_hourly_demand.csv

Classification: SYNTHETIC - PYTHON SIMULATION

Purpose: 2,160 rows x 10 columns. Hourly profile for 3 eligible station profiles for 2-31 December 2025: 3 stations x 30 days x 24 hours.

Columns: station_id, datetime, date, hour, hourly_cng_demand_kg, day_of_week, is_weekend, simulation_profile, data_type, generation_method

Generation logic: assumed morning, midday, and evening peak curves, profile-specific weights, random variation, and normalization so each station-day hourly sum reconciles to its synthetic daily total.

Why synthetic? Real hourly dispenser/sales records were not obtained; it demonstrates the hourly profile visualization and aggregation workflow.

Important limitation: Hourly demand is not observed demand, vehicle arrival rate, service time, queue length, or measured waiting time.

4. Metadata and real public context CSVs

optional/cng_infrastructure.csv

Classification: REAL PUBLIC CONTEXT - HISTORICAL / MIXED GEOGRAPHIC SCOPES

Purpose: 4 rows x 12 columns. PPAC figures retained with their original scope and dates.

Columns: record_id, indicator_name, indicator_value, unit, geographic_scope, reporting_period, publication_date, source_name, source_url, source_access_date, data_type, notes

Source websites: Petroleum Planning & Analysis Cell (PPAC), Government of India: December 2021 Ready Reckoner, Table 3.8, with CNG station counts as of 30 September 2021 - PPAC Ready Reckoner PDF; and PPAC Tracker for the India-wide CNG stations figure shown as of 31 August 2026 - PPAC Tracker.

Important limitation: The historic authorized-area boundaries (e.g., Ahmedabad City and Daskroi, Ahmedabad City/HPCL area, Ahmedabad District/EAAA) are not interchangeable with current Ahmedabad municipal limits. The national figure is not an Ahmedabad estimate.

data_sources.csv

Classification: PROVENANCE METADATA

Purpose: 18 rows x 13 columns listing sources actually accessed, sources reviewed but not successfully extracted, synthetic-generation method, geographic scope, dates, limitations, and attribution notes.

Columns: source_id, dataset_name, source_name, publisher, source_url, geographic_scope, reporting_period, access_date, data_origin, verification_status, licence_or_attribution, limitations, notes

Use: Audit where each public record or research lead came from, and distinguish a source used for a dataset from a website merely reviewed.

data_dictionary.csv

Classification: SCHEMA DOCUMENTATION

Purpose: 103 rows x 9 columns. One row per documented column across the nine CSV files, including each field name, data type, units, meaning, example, allowed values, required status, and origin.

Columns: file_name, column_name, data_type, unit, description, example_value, allowed_values_or_constraints, required_or_optional, data_origin

Use: The field reference when coding CSV readers, API schemas, and database models.

data_gaps.csv

Classification: RESEARCH GAP REGISTER

Purpose: 8 rows x 8 columns documenting the information sought, sources attempted, result, reason unavailable/unsuitable, project impact, and recommended next step.

Columns: dataset_or_information_needed, attempted_sources, source_urls, information_sought, result, reason_unavailable_or_unsuitable, effect_on_project, recommended_next_step

Important gaps captured: no Ahmedabad-specific Vahan export, no structured Ahmedabad station list from the Adani locator, no processed dated municipal boundary, no authorized station-level sales/supply, and no complete verified cross-operator inventory.

5. Python scripts and how to run the package

generate_synthetic_data.py

Classification: GENERATION SCRIPT - NOT A FORECASTING MODEL

Purpose: Reads cng_stations.csv and writes or refreshes synthetic_daily_sales.csv, synthetic_supply.csv, and synthetic_hourly_demand.csv.

What it does: fixes the random seed; uses only station rows marked in_study_area=yes; creates the 2024-2025 daily simulation; creates matching planned supply; creates hourly curves for up to three profiles; and runs key checks. It does not edit cng_stations.csv.

Dependencies: Python 3, pandas, NumPy.

Run from the extracted folder:

```
python generate_synthetic_data.py
```

validate_package.py

Classification: VALIDATION SCRIPT - INTEGRITY CHECKS

Purpose: Checks the package files, station master schema and IDs, unapproved coordinate status, daily date coverage, supply key matching, hourly totals, provenance records, and data dictionary. It writes validation_report.md.

Dependencies: Python 3, pandas, NumPy.

Run from the extracted folder:

```
python validate_package.py
```

A successful technical validation means that the file relationships and simulation checks passed. It does not mean the station list is complete, station coordinates are verified, or synthetic data is operationally accurate.

Validation snapshot (29 October 2026)

Check	Result
Daily sales and supply	8,041 rows each; same station-date keys; no duplicate keys or negative values.
Hourly simulation	2,160 rows; 24 hourly rows per station-day; sums reconcile to daily totals with maximum difference 0.00000 kg.
Station coordinates	0 approved coordinates in station master; all 8 candidate points remain unapproved.
Baseline metrics	28-day rolling mean, chronological synthetic test period 2025-10-03 to 2025-12-31; 990 station-days; MAE 150.55 kg/day, RMSE 189.69 kg/day, sMAPE 11.41%.

Baseline results are synthetic-demo metrics only. They do not show the model's expected performance on real Ahmedabad demand.

6. Websites researched and what was actually used

This list follows data_sources.csv. "Used" means a specific record or context indicator was extracted. "Reviewed only" means the source was accessed but did not yield a suitable dataset for this package.

Website / publisher	URL	Status in this package
Gujarat Energy Limited (formerly Gujarat Gas Limited)	CNG Stations directory	Used for 2 Chandkheda station name/address rows; no coordinates.
Hindustan Petroleum Corporation Limited (HPCL)	CNG outlet directory	Used for 10 Ahmedabad-address entries; address-level only.
CNGbunk.com	Secondary directory	Used only to populate 8 candidate coordinate leads; not approved for map pins.
Petroleum Planning & Analysis Cell (PPAC)	Ready Reckoner	Used for historical CNG station indicators with original geographic labels.
PPAC	CNG station tracker	Used for India-wide tracker figure only; not Ahmedabad-specific.
Ministry of Road Transport & Highways / NIC - Vahan	Vahan dashboard	Reviewed; no reusable Ahmedabad-specific registration export was retrieved.
Adani Total Gas Limited	CNG locator	Reviewed; dynamic locator did not yield a structured Ahmedabad station table.
WhereFUEL / OpenStreetMap contributors	Ahmedabad mapped index	Reviewed as a secondary index; not treated as exhaustive or used to fill coordinates.
DataMeet - Spatial Data of Municipalities	Ahmedabad boundary landing page	Reviewed; ward GeoJSON advertised but not downloaded/processed for point-in-polygon validation.

Attribution and reuse

The source inventory retains the URLs and caveats. Confirm reuse terms for operator directories and secondary station pages before redistributing derived records. If OpenStreetMap data is reused, follow its attribution and ODbL requirements. The coordinate candidate file is a review queue, not authoritative geospatial data.

7. What is still missing before full development?

Gap	What must be done
Verified station coordinates and boundary	Obtain coordinates from authoritative/operator sources; independently match names and addresses; process a suitable dated Ahmedabad municipal boundary; separate uncertain fringe locations.
More complete cross-operator station coverage	Review Gujarat Energy/Gujarat Gas, HPCL, Adani Total Gas, and other applicable operators. Do not infer nonexistence from absence in one directory.
Actual station-level operating history	Request authorized daily/hourly sales in kg and planned supply/delivery records from Gujarat Gas or other operators. Keep data dictionary, station IDs, time zone, units and data quality notes consistent.
Ahmedabad-specific vehicle registrations	If available, export Vahan data by the correct RTO/fuel/vehicle class/period and distinguish new registrations from cumulative registered vehicles.
Verified holidays and contextual indicators	Add sourced official holiday dates only after checking a reliable Gujarat calendar. Add traffic, accessibility, land use, fuel prices and EV trends only with defined geography, period, and source.

Important terms

Term	Meaning
Synthetic data	Artificial values made by a documented algorithm for testing or demonstration. These are not measured values.
AI-generated data	Values directly invented or generated by an AI model. The package time series are not this: their numeric values are generated by Python simulation rules.
Forecast demand	A model estimate for a future date. It is not the same as historical sales.
Planned available supply	An assumed amount available for a day in the simulation. It is not actual stock or a company delivery record.
Potential planning shortfall	max(0, forecast demand - planned available supply). This is an alert for human review, not proof of a stockout.
Net additional network demand	Estimated new-station demand minus estimated demand diverted from existing stations. All assumptions must be adjustable and visible.

Final decision

Ready now: CSV ingestion, synthetic forecasting experiments, supply-alert UI testing, hourly-profile charts, and validation tooling. Not ready yet: verified production map pins, distance-based ranking, evidence-backed location-impact estimates, or real-world forecasting claims. Close coordinate and real-data gaps before using the results for actual investment decisions.

Package reference: CNG_SmartPlan_Data.zip. For column-by-column details, use data_dictionary.csv. For the full source log and limitations, use data_sources.csv, data_gaps.csv, and validation_report.md.